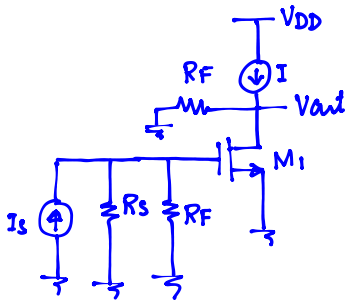
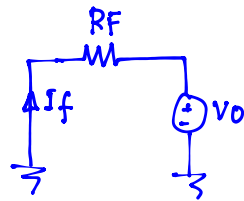


## Quiz-I (PART B) Answerkey



$$V_o = I_s (R_s \parallel R_F) \times -g_m (r_o \parallel R_F)$$

$$\therefore A = \frac{V_o}{I_s} = -g_m (r_o \parallel R_F) (R_s \parallel R_F)$$



$$\beta = \frac{I_f}{V_o} = -\frac{1}{R_F}$$

$$A\beta = \frac{g_m (r_o \parallel R_F) (R_s \parallel R_F)}{R_F}$$

$$A_f = \frac{A}{1 + \beta A} = \frac{-g_m (r_o \parallel R_F) (R_s \parallel R_F)}{1 + \frac{g_m (r_o \parallel R_F) (R_s \parallel R_F)}{R_F}}$$

$$R_i = R_F, \quad R_{i,f} = \frac{R_F}{1 + \beta A}$$

$$R_o = R_F \parallel r_o, \quad R_{o,f} = \frac{R_F \parallel r_o}{1 + \beta A}$$

For given values

$$A = -2\text{m} (40\text{k} \parallel 20\text{k}) (2\text{k} \parallel 20\text{k}) = -48.48 \text{ k}\Omega$$

$$\beta = \frac{1}{R_F} = \frac{-1}{20\text{k}} = -0.05 \text{ mA/V}$$

$$A\beta = -48.48 \text{ k}\Omega \times -0.05 \text{ mA/V} = 2.424$$

$$A_f = \frac{A}{1 + \beta A} = \frac{-48.48}{1 + 2.424} = -14.16 \text{ k}\Omega$$

$$R_i = R_F = 20 \text{ k}\Omega$$

$$R_{in} = R_F \parallel R_s = 20\text{k} \parallel 2\text{k} = 1.82 \text{ k}\Omega$$

$$R_{in,f} = \frac{R_{in}}{1 + \beta A} = \frac{1.82 \text{ k}\Omega}{1 + 2.424} = 0.53 \text{ k}\Omega$$

$$R_{i,f} = \frac{1}{\frac{1}{R_{in,f}} - \frac{1}{R_s}} = \frac{1}{\frac{1}{0.53\text{k}} - \frac{1}{2\text{k}}} = 721.1 \Omega$$

$$R_o = R_F \parallel r_o = 20\text{k} \parallel 40\text{k} = 13.33 \text{ k}\Omega$$

$$R_{o,f} = \frac{R_o}{1 + \beta A} = \frac{13.33 \text{ k}\Omega}{1 + 2.424} = 3.89 \text{ k}\Omega$$